Ali Husain

Software engineer and product designer

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gallowforge.github.io
github.com/gallowforge
Remote, UTC+3 · Open to remote work

I build complete systems, from the interface to the infrastructure.

IN NUMBERS

11
public repositories, each one specified, tested, documented and released

4,255
automated tests across them, run on every change

15
services in production on a platform I designed and operate

SELECTED WORK

Ballast

LIVE APP
OPEN SOURCE

Personal finance that turns savings into a salary that lasts.
Designed and built end to end, offline-first, with no account and no server. Says what is safe to spend today, draws how long the money lasts, and reads bank statements on the device, checking every row against the running balance.

599 automated tests, unit and end-to-end
40 currencies, in exact whole-number maths

gallowforge.github.io/ballast
github.com/gallowforge/ballast

Ethereal

LIVE APP
OPEN SOURCE

A calm coach for training, food and the shape of the day.
An installable app that plans the day, guides workouts and tracks nutrition, with no account, no ads and no trackers. Product design, front end and delivery, end to end, built to a WCAG 2.1 AA target with Lighthouse at 100.

397 automated tests, run on every push
1,474 foods in a searchable offline database

gallowforge.github.io/Ethereal
github.com/gallowforge/Ethereal

Holdfast

LIVE DEMO
OPEN SOURCE

A memory for coding agents in which every fact carries its own proof.
Each fact is stored with a check that proves it, run again before the agent is told. What has stopped being true comes back marked broken, with the evidence. One Go binary.

1,651 automated tests, on Linux and macOS
10 kinds of check, from a file to a live endpoint

gallowforge.github.io/holdfast
github.com/gallowforge/holdfast

Seaward

LIVE DEMO
OPEN SOURCE

Shows which ports on a server can really be reached from outside, and the rule that decides each one.
Reads the rules the kernel has loaded and follows a packet through them, so a port Docker published past the firewall's list is found and explained. One Go binary.

964 automated tests, run on every change
1,950 real packets checked against the kernel

gallowforge.github.io/seaward
github.com/gallowforge/seaward

Vesper

IN PRODUCTION
OPEN SOURCE

A read-only operations console that answers one question: does anything need you?
One Go binary with the interface inside, read-only by construction. When it cannot confirm something it says so instead of showing an all-clear. Now at version 2.2.0.

7.4 MB one Go binary, about 27 MB of memory
191 tests, 34 end to end under real systemd

gallowforge.github.io/vesper
github.com/gallowforge/vesper

homelab

DESIGNER AND SOLE OPERATOR

Fifteen services on one small machine, run as code, with backups that are proven every week.
Remote access with zero port forwards. Every backup set is verified nightly, copied to three machines and restored in a weekly drill. 410 commits, about 4,100 lines of runbooks.

15 containers, with 21 scheduled services
77 alert conditions and 13 uptime probes

github.com/gallowforge/homelab

TOOLS — Each at github.com/gallowforge/name

agent-guardrails	Agents that judge work they did not write, and a warning before a destructive command.	36 tests
evalbench	Finds the cheapest language model that clears your bar, with checks and blind ratings.	291 tests

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CAPABILITIES

		Where to see it
Product design	Interfaces designed for phone and desktop together, accessible by default, and measured with real devices and automated audits.	Ballast, Ethereal, Vesper Ethereal to a WCAG 2.1 AA target, with Lighthouse at 100
Front end	Installable apps that work offline, with no account and no server behind them, and a live operations console streamed to the browser.	Ballast, Ethereal, Vesper 599 and 397 automated tests on the two apps
Back end	Services and command-line tools in Python, Go, TypeScript, Bash, SQL and C/C++, with the documentation that ships with them.	Holdfast, Seaward, Vesper, Evalbench Three Go binaries, each with no third-party Go dependencies
AI evaluation and agent workflows	Model evaluation with automatic checks and blind human rating. Agent memory that is proved again before use, and reviewer and verifier agents that judge work they did not write.	Holdfast, Evalbench, agent-guardrails One build ran as nine coordinated agents, a platform audit as thirteen
Infrastructure and operations	Linux, containers, systemd, CI/CD and infrastructure as code, with monitoring and incident response. Backups are proven by a restore drill every week.	homelab, Vesper 410 commits, 21 scheduled services, about 4,100 lines of runbooks
Security practice	DNS, TLS, mesh VPN, firewalls, service sandboxing, intrusion prevention, encryption and secrets handling. Zero port forwards and one audited public endpoint.	Seaward, homelab, Vesper, Evalbench 7-node mesh VPN; code tests run in a sandbox with no network
Video pipeline	A documented production method for painted films made with AI image and video models, with post-production built in code, from beat-matched editing to 4K mastering on GPU.	filmforge 9 skills and 7 tested tools; every encode verified

HOW I WORK

Specified before it is built, tested before it ships, documented so it runs without me.

Prove it works A backup isn't done until it's restored; a fix isn't done until it's tested from the user's side.

Write it down Every system ships with a runbook someone else could follow.

Learn by shipping Linux, Go, AI agents, post-production: each learned by building something real.

Fewer things, done properly Simple setups that stay up beat clever ones that break.

Tested in public

github.com/gallowforge/	tests
holdfast	1,651
seaward	964
ballast	599
Ethereal	397
evalbench	291
vesper	191
homelab	43
filmforge	40
agent-guardrails	36
aih	31
bwtidy	12
11 public repositories	**4,255**

EDUCATION

Computer Science and Engineering

CONTACT

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